

1.1 Physical Quantities & Units

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Medium

Time allowed: 80

Score: /66

Percentage: /100

Question 1a

Define the following quantities and state their SI base units

(i)

velocity

[2]

(ii)

acceleration

[2]

(iii)

density

[2]

[6 marks]

Question 1b

The pressure P due to a liquid of density ρ is related to the depth h by the expression

$$P = \rho gh$$

where g is the acceleration of free fall.

Use this expression to determine the derived units of pressure P .

[2 marks]

Question 1c

The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

where γ is a constant.

Show that the constant γ has no unit.

[3 marks]

Question 2a

Estimate the speed of an apple as it hits the ground after falling from a tree.

[2 marks]

Question 2b

Estimate the number of kilowatt-hours used by a lightbulb if switched on for a whole day.

[2 marks]

Question 2c

Estimate the density, in g cm^{-3} , of the head of an adult person.

[4 marks]**Question 2d**

The radius of the Earth is approximately $6.4 \times 10^6 \text{ m}$.

Estimate the number of people it would take to circle the Earth holding hands.

[4 marks]**Question 3a**

Derive the SI base unit of force.

[2 marks]

Question 3b

As a spherical object of radius r moves through the air at speed v , it experiences a force F due to air resistance.

The resistive force F is given by

$$F = krv$$

Where k is a constant.

Derive the SI base unit of the constant k .

[2 marks]

Question 3c

The relationship can be used to model a spherical raindrop falling vertically in the air.

Fig 1.1 shows the forces acting on the raindrop as it accelerates downwards.

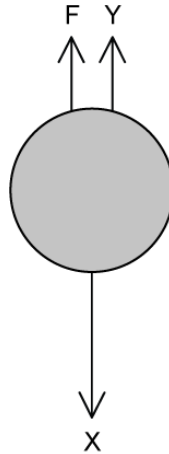


Fig. 1.1

The raindrop has a radius of 1.5 mm and falls at a velocity of 3.5 m s^{-1} . The constant k has a magnitude of 0.0124 in air.

Other than the resistive force F described in (b), two other forces, X and Y , act on the raindrop as it falls through the air.

- (i)
State the names of forces X and Y .

[2]

- (ii)
Calculate the magnitude of the resistive force F in μN .

[2]

- (iii)
Force X has a value of $137 \mu\text{N}$ and force Y has a value of $0.18 \mu\text{N}$.

Using Fig. 1.1, determine the magnitude and direction of the resultant force on the raindrop.

[3]

[7 marks]

Question 3d

On a windy day, the resultant force on the raindrop is $105\ \mu\text{N}$ at an angle of θ to the vertical as shown in Fig. 1.2.

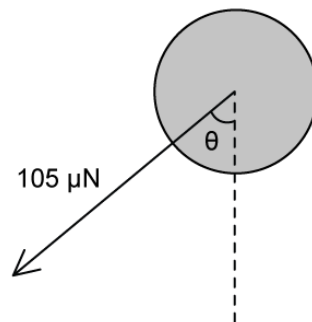


Fig. 1.2

Use your answer to (c)iii to show that the angle θ between the resultant and the vertical is about 47° .

[3 marks]

Question 4a

(i)

State the difference between a scalar quantity and a vector quantity.

[1]

(ii)

Underline all the vectors in the list below.

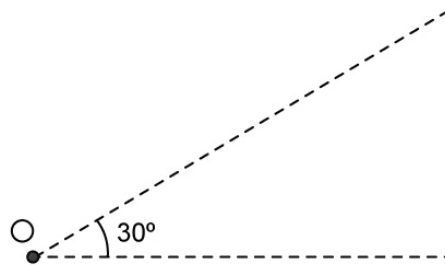
power length density velocity momentum

[1]

[2 marks]**Question 4b**

Two gusts of wind exert forces of magnitude 5.5 N and 11.9 N on a wind turbine at a point O. Both forces act away from point O and the angle between them is 30° .

Fig. 1.1 shows two lines at an angle of 30° to one another.

**Fig. 1.1**

On Fig. 1.1, draw a vector diagram to determine the magnitude of the resultant of the two forces.

[4 marks]

Question 4c

Fig 1.2 shows a turbine that is used to generate electrical power P from wind travelling with velocity v .

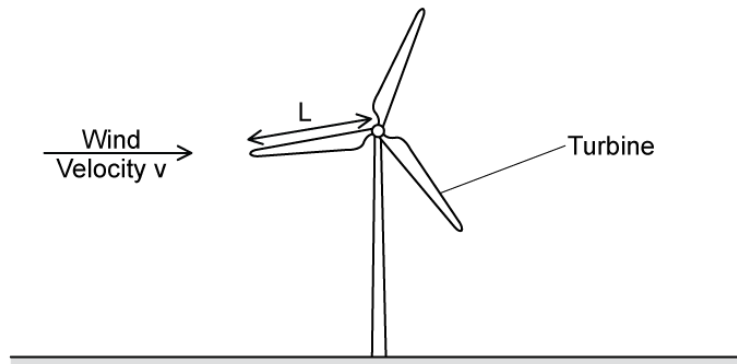


Fig. 1.2

The useful power output P of the turbine is given by

$$P = C_p L^2 \rho v^3$$

Where L is the length of each blade, ρ is the density of air, v is the wind velocity and C_p is a constant.

Show that the constant C_p has no units.

[5 marks]

Question 4d

A local council is looking into building wind turbines to supply power to 2500 households in a village. The average energy consumption of a household in the village is 3000 kWh a year.

Fig. 1.3. shows an image from a catalogue detailing the different sizes and power outputs of available wind turbines.

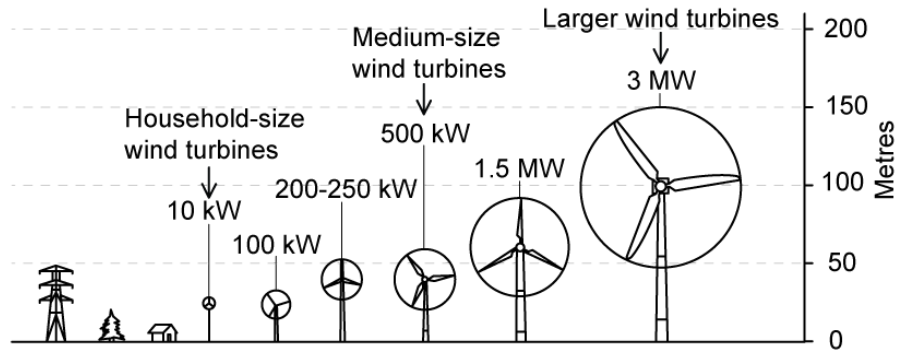


Fig. 1.3

They decide to build 8 of the same wind turbine to supply the entire village with electricity.

(i)

Determine which of the wind turbines in Fig. 1.3. the council decided to build.

[3]

(ii)

The chosen turbine delivers the power specified at a C_P value of 0.45. This value can only be achieved above certain wind velocities.

Estimate the average wind velocity needed to power the village for a year. The density of air is 1.3 kg m^{-3} .

[2]

[5 marks]

Question 5a

Fig. 1.1 shows a 2 tonne wrecking ball being used on a demolition site.

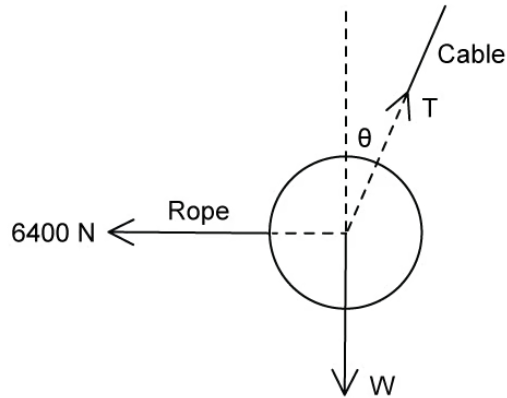


Fig. 1.1

The ball is suspended from a cable which makes an angle θ with the vertical. The ball is pulled into the position shown by a rope that is kept horizontal. The tension in the rope is 6400 N .

Write expressions for tension T using

- (i)
the vertical component of the tension in the cable
- (ii)
the horizontal component of the tension in the cable.

[2]

[2]

[4 marks]

Question 5b

Determine

(i)
the angle θ the cable makes with the vertical

[2]

(ii)
the magnitude of the tension T in the cable

[2]

[4 marks]

Question 5c

On the same demolition site, there is a uniform beam supported by two light cables, **AB** and **AC**, which are attached to a single steel cable from a crane, as shown in Fig. 1.2.

The beam is stationary and in equilibrium.

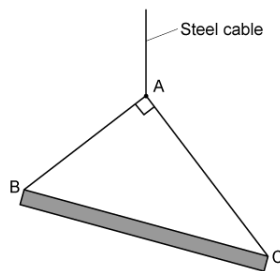


Fig. 1.2

Draw the vector triangle for this situation labelling the tension in both cables and the weight of the beam.

[3 marks]

Question 5d

The tension in the cable **AB** is 9 N and the tension in the cable **AC** is 12 N.

Calculate the resultant force required in the beam **BC** to keep the system in equilibrium.

[2 marks]

Model Answers: Medium

1a

(a)

(i) Velocity is defined as

- Change in displacement per unit time; [1 mark]
- SI base unit = m s^{-1} [1 mark]

(ii) Acceleration is defined as

- Change in velocity per unit time; [1 mark]
- SI base unit = m s^{-2} [1 mark]

(iii) Density is defined as

- Mass per unit volume; [1 mark]
- SI base unit = kg m^{-3} [1 mark]

[Total: 6 marks]

1b

(b) Use the expression to determine the derived unit for pressure:

- $P = \rho gh$
- Unit of density $[\rho] = \text{kg m}^{-3}$
- Unit of acceleration $[g] = \text{m s}^{-2}$
- Unit of height $[h] = \text{m}$

Determine the unit of pressure:

- $[P] = [\rho][g][h]$

- $[P] = (\text{kg m}^{-3}) \times (\text{m s}^{-2}) \times (\text{m})$ [1 mark]
- $[P] = \text{kg m}^{-1} \text{s}^{-2}$ [1 mark]

[Total: 2 marks]

1c

(c) Show that the constant γ has no unit:

- $v = \sqrt{\frac{\gamma P}{\rho}}$
- Unit of density $[\rho] = \text{kg m}^{-3}$
- Unit of velocity $[v] = \text{m s}^{-1}$
- Unit of pressure $[P] = \text{kg m}^{-1} \text{s}^{-2}$

Determine the units on both sides of the expression:

- $[v] = \sqrt{\frac{[\gamma][P]}{[\rho]}}$
- LHS: $[v] = \text{m s}^{-1}$ [1 mark]
- RHS: $\sqrt{\frac{[P]}{[\rho]}} = \sqrt{\frac{\text{kg m}^{-1} \text{s}^{-2}}{\text{kg m}^{-3}}}$ [1 mark]
- RHS: $\sqrt{\frac{[P]}{[\rho]}} = \sqrt{\text{m}^{-1} \text{m}^3 \text{s}^{-2}} = \sqrt{\text{m}^2 \text{s}^{-2}} = \text{m s}^{-1}$ [1 mark]
- LHS = RHS, so γ has no unit

[Total: 3 marks]

2a

(a) Estimate the speed of an apple as it hits the ground after falling from a tree:

- Height of apple on a tree, $h \approx 5 \text{ m}$ [1 mark]

Acceptable range for $h = 2$ to 20 m

Use energy conservation to estimate the speed:

- Change in GPE = Gain in KE
- $mgh = \frac{1}{2}mv^2$
- $v^2 = 2gh \Rightarrow v = \sqrt{2gh}$
- $v \approx \sqrt{2 \times 9.81 \times 5}$
- $v \approx 9.91 \approx 10 \text{ m s}^{-1}$ [1 mark]

Acceptable range for $v = 6 \text{ to } 20 \text{ m s}^{-1}$

[Total: 2 marks]

2b

(b) Estimate the number of kilowatt-hours used by a lightbulb if switched on for a whole day:

- Power output of a lightbulb $\approx 70 \text{ W}$ [1 mark]

Acceptable range for power = 20 to 100 W

Convert from W to kWh in 24 hours:

- $P = \frac{E}{t} \Rightarrow E = Pt$
- Kilowatt-hours is a unit of energy, so $E(kWh) = \frac{P(W) \times t(hr)}{1000}$
- $E(kWh) \approx \frac{70 \times 24}{1000}$
- $E(kWh) \approx 1.68 \approx 1.7 \text{ kWh per day}$ [1 mark]

Acceptable range for $E(kWh) = 0.5 \text{ to } 2.4 \text{ kWh}$

[Total: 2 marks]

2c

(c) Estimate the density, in g cm^{-3} , of the head of an adult person:

- Radius of a head, $r \approx 10 \text{ cm}$ [1 mark]

Acceptable range for radius = 7 to 12 cm

- Mass of a head, $m \approx 5 \text{ kg} = 5000 \text{ g}$ [1 mark]

Acceptable range for mass = 1 to 10 kg (1000 to 10 000 g)

Use the equation for volume of a sphere with the density equation:

- Volume of an adult's head $\approx$ volume of a sphere

- Volume of a sphere: $V = \frac{4}{3} \pi r^3$

- Density: $\rho = \frac{m}{V} = \frac{m}{\frac{4}{3} \pi r^3}$

- $\rho = \frac{5000}{\frac{4}{3} \pi \times (10)^3}$ [1 mark]

- $\rho = 1.194 = 1.2 \text{ g cm}^{-3}$ [1 mark]

Acceptable range for density = 0.1 to 7.0 g cm^{-3}

[Total: 4 marks]

When asked a question, such as this, where you likely have no reference to make a reasonable estimate for something, like the mass of a person's head, think about reasonable approximations you could make

For example, if the mass of an adult person is about 70 kg, you could make a reasonable assumption that their head might make up about 10% of this mass, which would be 7 kg

It's important that you communicate this in your answer so the examiner can see your lines of thought, and even if you are outside of the acceptable range, you could still earn extra marks for method and reasoning

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2d

(d) Estimate the number of people it would take to circle the Earth holding hands:

- Radius of Earth, $r = 6.4 \times 10^6 \text{ m}$
- Arm span $\approx 2 \text{ m}$ [1 mark]

Acceptable range for arm span estimate = 1 to 3 m

Calculate the circumference of Earth:

- Circumference of a circle = $2\pi r$
- Circumference of Earth = $2\pi \times (6.4 \times 10^6) = 4 \times 10^7 \text{ m}$ [1 mark]

Calculate how many people would be needed:

- Each person spans approximately 2 m
- Number of people $\approx \frac{\text{circumference of Earth}}{\text{arm span}}$
- Number of people $\approx \frac{4 \times 10^7}{2}$ [1 mark]
- Number of people $\approx 2 \times 10^7$ [1 mark]

Acceptable range for number of people estimate = 1.3×10^7 to 4×10^7

[Total: 4 marks]

If you have approximated the arm span of a person to a different value than 2 m, that's ok, as long as it is a **reasonable** approximation. You may have said, 1.5 m or somewhere in between, and that's ok. You would still get the marks. But, if you approximated it to 5 m, this is **not** a reasonable estimation and you would not get the first mark. However, you would still get the other two marks if your calculations were correct, no matter what number you used.

Questions that ask you to **estimate** may not always contain an unknown value that you need to approximate, like arm span in this question. They may just use inexact values such as the diameter of the Earth, or the mass of a proton as 1 atomic mass unit. Don't let this throw you, the calculation is just the same as it would be if you were using exact values. If you do approximate something, like the hand span, just remember to use the 'approximately equal to' sign $\approx$

3a

(a) Derive the SI base unit of force:

- Force: $F = ma$ [1 mark]
- Unit of mass: kg
- Unit of acceleration: m s^{-2}

Determine the unit of force:

- $[F] = [m][a]$
- $[F] = \text{kg m s}^{-2}$ [1 mark]

[Total: 2 marks]

3b

(b) Derive the SI base unit of the constant k :

- $F = krv$
- Unit of force $[F] = \text{kg m s}^{-2}$
- Unit of radius $[r] = \text{m}$
- Unit of speed $[v] = \text{m s}^{-1}$

Determine the units on both sides of the expression:

- $[F] = [k][r][v] \Rightarrow [k] = \frac{[F]}{[r][v]}$
- $[k] = \frac{\text{kg m s}^{-2}}{(\text{m})(\text{m s}^{-1})}$ [1 mark]
- $[k] = \text{kg m}^{-1} \text{s}^{-1}$ [1 mark]

[Total: 2 marks]

3c

(c)

(i) The names of forces X and Y are:

- Force X = Weight; [1 mark]
- Force Y = Upthrust; [1 mark]

(ii) Calculate the magnitude of the resistive force F :

List the known quantities:

- Radius, $r = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$
- Velocity, $v = 3.5 \text{ m s}^{-1}$
- Constant, $k = 0.0124 \text{ (kg m}^{-1} \text{ s}^{-1})$

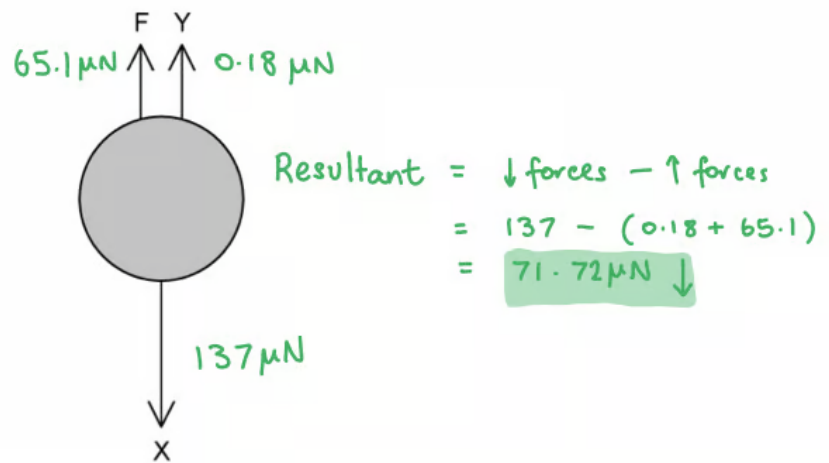
Calculate the resistive force F :

- Resistive force: $F = krv$
- $F = 0.0124 \times (1.5 \times 10^{-3}) \times 3.5 = 6.51 \times 10^{-5} \text{ N}$ [1 mark]

Convert to μN (10^{-6} N):

- $F = 6.51 \times 10^{-5} \text{ N} \rightarrow 65.1 \times 10^{-6} \text{ N}$
- $F = 65.1 \mu\text{N}$ [1 mark]

(iii) Determine the magnitude and direction of the resultant force on the raindrop:



- (From Fig 1.1) Resultant force = $X - Y - F$ **OR** $X - (Y + F)$ [1 mark]
- Resultant force = $137 - 0.18 - 65.1 = 71.72 \mu\text{N}$
- Magnitude of the resultant = $72 \mu\text{N}$ (2 s.f.) [1 mark]
- Direction of the resultant = downwards / in the direction of X [1 mark]

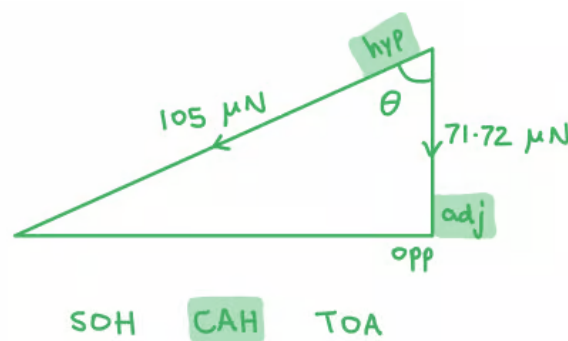
[Total: 7 marks]

3d

(d) Show that the angle θ between the resultant and the vertical is about 47° :

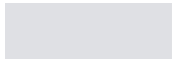
- Resultant force (down) = $71.72 \mu\text{N}$
- Resultant force (windy day) = $105 \mu\text{N}$

Draw a vector triangle and use trigonometry to find the angle θ :



- $\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{71.72}{105}$ [1 mark]

- $\theta = \cos^{-1} \left(\frac{71.72}{105} \right)$ [1 mark]



- (From Fig 1.1) Resultant force = $X - Y - F$ **OR** $X - (Y + F)$ [1 mark]
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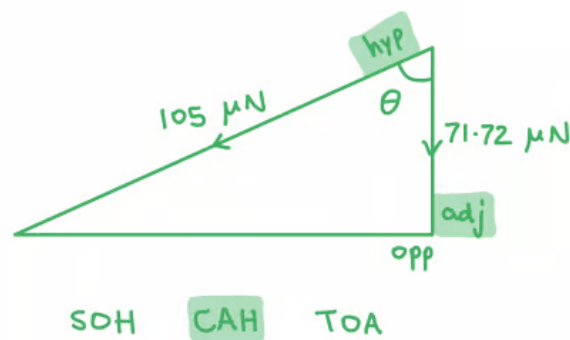
[Total: 7 marks]

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Draw a vector triangle and use trigonometry to find the angle θ :



- $\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{71.72}{105}$ [1 mark]
- $\theta = \cos^{-1} \left(\frac{71.72}{105} \right)$ [1 mark]
- $\theta = 46.918 = 46.9^\circ$ [1 mark]

[Total: 3 marks]

4a

(a)

(i) The difference between scalar and vector quantities is:

- Scalars have magnitude only
AND
- Vectors have both magnitude and direction; [1 mark]

(ii) The vector quantities are:

- Velocity and momentum only; [1 mark]
- Power, length and density are scalars

[Total: 2 marks]

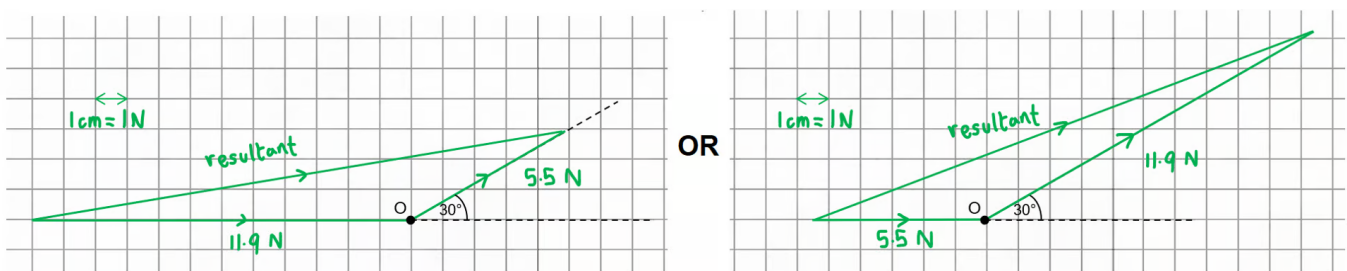
4b

(b) Draw a vector diagram of the two forces and determine the resultant:

List the known quantities:

- Force from wind 1 = 11.9 N
- Force from wind 2 = 5.5 N
- Angle between the forces = 30°

Construct a vector triangle:



- Forces drawn end to tail with arrows and labels (either way around); [1 mark]
- Angle between the forces is used correctly; [1 mark]
- Resultant drawn correctly to form a triangle; [1 mark]

Determine the magnitude of the resultant:

- Resultant = 17.1 N (at least 2 sig. fig) [1 mark]

Acceptable range for resultant = 16.8 to 17.4 N

Calculated answer with no sketch = no marks

[Total: 4 marks]

The best approach to a vector drawing question is to follow these steps:

- First, draw one of the vectors, either is fine
- Next, draw the next vector with its **tail** at the **head** of the first vector
- Then, draw the resultant from the **tail of the first** vector to the **head of the second** vector

In this example, we have used gridlines to help to show the scale, however, in an exam you will be expected to draw on plain paper with just your ruler and protractor, so make sure you get lots of practice with these!

4c

(c) Show that the constant C_P has no units:

- $P = C_P L^2 \rho v^3$
- Unit of length $[L] = \text{m}$
- Unit of density $[\rho] = \frac{[m]}{[V]} = \text{kg m}^{-3}$ [1 mark]
- Unit of velocity $[v] = \text{m s}^{-1}$

Determine the SI unit of power:

- Unit of power $[P] = \frac{[E]}{[t]} = \frac{[m][v]^2}{[t]} = \frac{(\text{kg})(\text{m s}^{-1})^2}{\text{s}}$ [1 mark]
- $[P] = \text{kg m}^2 \text{s}^{-2} \text{s}^{-3} = \text{kg m}^2 \text{s}^{-3}$ [1 mark]

Determine the units on both sides of the expression:

- $[P] = [C_P][L]^2[\rho][v]^3$
- LHS: $[P] = \text{kg m}^2 \text{s}^{-3}$
- RHS: $[L]^2[\rho][v]^3 = (\text{m})^2 (\text{kg m}^{-3}) (\text{m s}^{-1})^3$ [1 mark]

- RHS: $[L]^2[\rho][v]^3 = \text{kg m}^2 \text{m}^{-3} \text{m}^3 \text{s}^{-1} = \text{kg m}^2 \text{s}^{-1}$ [1 mark]
- LHS = RHS, so C_P has no unit

[Total: 5 marks]

4d

(d)

List the known quantities:

- Number of households = 2500
- Average energy consumption per household per year = 3000 kWh
- Number of turbines = 8
- Constant, $C_P = 0.45$
- Density of air, $\rho = 1.3 \text{ kg m}^{-3}$

(i) Determine the power rating of the wind turbine the council decided to build:

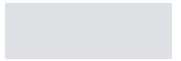
- Total energy (kWh) required by the village in a year: $2500 \times 3000 = 7\,500\,000 \text{ kWh}$
- Total energy required per day: $\frac{7\,500\,000}{365} = 20\,548 \text{ kWh}$
- Power output required: $\frac{20\,548}{24} = 856 \text{ kW}$ [1 mark]
- Power output required per turbine: $\frac{856}{8} = 107 \text{ kW}$ [1 mark]
- The council chose the 100 kW turbine; [1 mark]

(ii) Estimate the average wind velocity required to power the entire village:

- (From Fig 1.3) there are about $2\frac{3}{4}$ 100 kW turbine diameters in 50 m
- So, the diameter of the 1 turbine $\approx \frac{50}{2.75} \approx 18 \text{ m}$

- The length of a blade $L \approx 9 \text{ m}$ [1 mark]

Acceptable range for blade length $L = 7 \text{ to } 11 \text{ m}$



- RHS: $[L]^2[\rho][v]^3 = \text{kg m}^2 \text{m}^{-3} \text{m}^3 \text{s}^{-1} = \text{kg m}^2 \text{s}^{-3}$ [1 mark]
- LHS = RHS, so C_P has no unit

[Total: 5 marks]

4d

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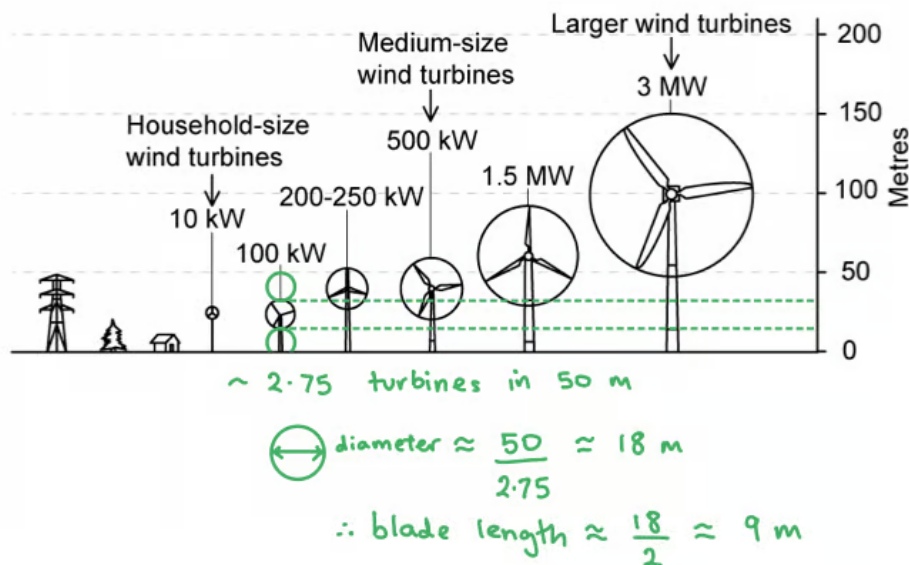
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- The length of a blade $L \approx 9 \text{ m}$ [1 mark]

Acceptable range for blade length $L = 7 \text{ to } 11 \text{ m}$



Determine the average wind velocity:

- $P = C_p L^2 \rho v^3 \Rightarrow v = \sqrt[3]{\frac{P}{C_p L^2 \rho}}$
- $v \approx \sqrt[3]{\frac{100\,000}{0.45 \times 9^2 \times 1.3}} \approx 13 \text{ m s}^{-1}$ [1 mark]
- The wind velocity must average at least 13 m s^{-1} in order for the turbines to maintain a power output of 100 kW

[Total: 5 marks]

5a

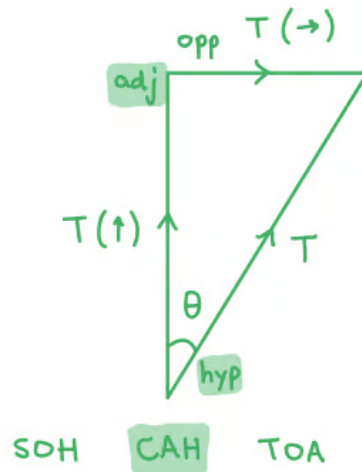
(a)

List the known quantities:

- Mass of the wrecking ball, $m = 2 \text{ tonne} = 2000 \text{ kg}$
- Tension in horizontal rope = 6400 N

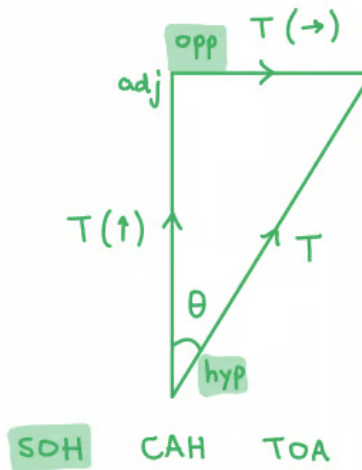
(i) Write an expression for the vertical component of the tension in the cable:

- Weight: $W = mg$
- $W = 2000 \times 9.81 = 19\,620\text{ N}$



- Resolving vertically: $T(\uparrow) = T \cos \theta = W$ [1 mark]
- $T(\uparrow) = \frac{W}{\cos \theta} = \frac{19\,620}{\cos \theta}$ [1 mark]

(ii) Write an expression for the horizontal component of the tension in the cable:



- Resolving horizontally: $T(\rightarrow) = T \sin \theta = 6400\text{ N}$ [1 mark]
- $T(\rightarrow) = \frac{6400}{\sin \theta}$ [1 mark]

[Total: 4 marks]

5b

(b)

(i) Determine the angle θ the cable makes with the vertical:

- Equate the expressions for T : $\frac{19\,620}{\cos \theta} = \frac{6400}{\sin \theta}$
- Rearrange to get $\tan \theta$: $\frac{\sin \theta}{\cos \theta} = \tan \theta = \frac{6400}{19\,620}$ [1 mark]
- Use the inverse tan function: $\theta = \tan^{-1} \left(\frac{6400}{19\,620} \right) = 18.1^\circ$ [1 mark]

(ii) Determine the magnitude of the tension T in the cable:

- Use either expression for T : $T = \frac{19\,620}{\cos \theta}$ **OR** $T = \frac{6400}{\sin \theta}$
- Substitute in the angle: $T = \frac{19\,620}{\cos 18.1^\circ}$ [1 mark]
- Magnitude of tension, $T = 20\,637 = 21\,000 \text{ N (2 s.f.)}$ [1 mark]

[Total: 4 marks]

In part (ii), you could also get the same result by using Pythagoras:

$$T = \sqrt{19\,620^2 + 6400^2} = 20\,637 \text{ N}$$

5c

(c) The correct vector triangle is as follows:

(b)

(i) Determine the angle θ the cable makes with the vertical:

- Equate the expressions for T : $\frac{19\,620}{\cos \theta} = \frac{6400}{\sin \theta}$
- Rearrange to get $\tan \theta$: $\frac{\sin \theta}{\cos \theta} = \tan \theta = \frac{6400}{19\,620}$ [1 mark]
- Use the inverse tan function: $\theta = \tan^{-1} \left(\frac{6400}{19\,620} \right) = 18.1^\circ$ [1 mark]

(ii) Determine the magnitude of the tension T in the cable:

- Use either expression for T : $T = \frac{19\,620}{\cos \theta}$ **OR** $T = \frac{6400}{\sin \theta}$
- Substitute in the angle: $T = \frac{19\,620}{\cos 18.1^\circ}$ [1 mark]
- Magnitude of tension, $T = 20\,637 = 21\,000 \text{ N (2 s.f.)}$ [1 mark]

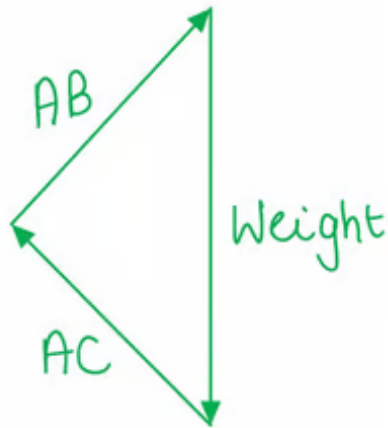
[Total: 4 marks]

In part (ii), you could also get the same result by using Pythagoras:

$$T = \sqrt{19\,620^2 + 6400^2} = 20\,637 \text{ N}$$

5c

(c) The correct vector triangle is as follows:



- All vectors are the correct size by eye; [1 mark]
- The vector triangle is closed and all vectors are joined tip to tail; [1 mark]
- All three vectors are correctly labelled; [1 mark]

[Total: 3 marks]

The key to answering this question correctly is realising that the weight of the beam will act vertically straight down. The other two vectors can then be slotted into the correct position.

5d

(d) Calculate the resultant force required to keep the system in equilibrium:

List the known quantities:

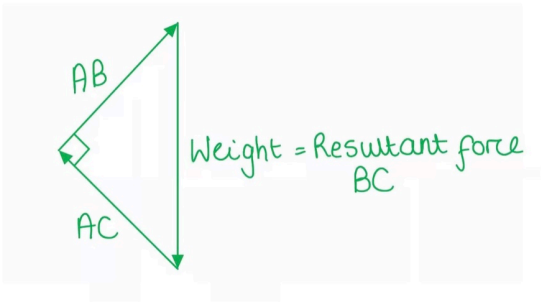
- Magnitude **AB** = 9 N
- Magnitude **AC** = 12 N

5d

(d) Calculate the resultant force required to keep the system in equilibrium:

List the known quantities:

- Magnitude **AB** = 9 N
- Magnitude **AC** = 12 N



- Force **BC** is the resultant force of **AB** and **AC** and is equal to the magnitude of the weight of the beam

Use Pythagoras to calculate resultant force:

- Resultant **BC** = $\sqrt{AB^2 + AC^2} = \sqrt{9^2 + 12^2}$ [1 mark]
- Resultant **BC** = 15 N [1 mark]

[Total: 2 marks]

Remember you only need to be able to resolve and calculate resultant components of vectors for right angle triangles. If you have an exam question about this you will most often need to use Pythagoras.